www.tntextbooks.in

Hence $x = \frac{\Delta x}{\Delta} = \frac{-44}{-11} = 4$, $y = \frac{\Delta y}{\Delta} = \frac{11}{-11} = -1$

$\therefore x = 4$ and $y = -1$

Answer checking:

Substituting in equation the values of x and y,

4 • 3(-1) • 1,

3(4) – 2(-1)• 14

Example: 12.9

Find the solution of the system of equations.

$5x_1 \bullet 3x_2 \bullet 30$

$6x_1 \bullet 2x_2 \bullet 8$

Solution:

The coefficient and the constant terms are given below for the equations

$$\Delta = \begin{vmatrix} 5 & 3 \\ 6 & -2 \end{vmatrix} = -10 - 18 = -28$$

$$\Delta x_1 = \begin{vmatrix} 30 & 3 \\ 8 & -2 \end{vmatrix} = -60 - 24 = -84$$

$$\Delta x_2 = \begin{vmatrix} 5 & 30 \\ 6 & 8 \end{vmatrix} = +40 - 180 = -140$$

$$\therefore x_1 = \frac{\Delta x_1}{\Delta} = \frac{-84}{-28} = 3$$

$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{-140}{-28} = 5$$

$\therefore x_1 \bullet 3, \quad x_2 \bullet 5$

Example: 12.10

Find the solution of the equation system

$7x_1 \bullet x_2 \bullet x_3 \bullet 0$

$10x_1 \bullet 2x_2 \bullet x_3 \bullet 8$

$6x_1 \bullet 3x_2 \bullet 2x_3 \bullet 7$

Solution:

The matrix form of the given equation is written as

$$\begin{bmatrix} 7 & -1 & -1 \\ 10 & -2 & 1 \\ 6 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 7 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 7 & -1 & -1 \\ 10 & -2 & +1 \\ 6 & 3 & -2 \end{vmatrix}$$

• •7(4• 3)• (• 1)(• 20• 6)• (• 1)(30• 12)

• •7(1) • •1(• 26) • •1(42)

• •7 • 26 • •42 • •• 61

$$= \begin{vmatrix} 0 & -1 & -1 \\ 8 & -2 & 1 \\ 7 & 3 & -2 \end{vmatrix}$$

• 0(4• 3)• (• 1)(• 16• 7) • •(• 1)(24 • •14)

• •0 • •1(• 23)• 1(38)

• •• 23 • •38 • •• 61

$$= \begin{vmatrix} 7 & 0 & -1 \\ 10 & 8 & 1 \\ 6 & 7 & -2 \end{vmatrix}$$

• •7(• 16• 7)• 0(• 20• 6)• (• 1)(70• 48)

• •7(• 23) • •0 • •1(22)

• •• 161• 22 • •• 183

$$= \begin{vmatrix} 7 & -1 & 0 \\ 10 & -2 & 8 \\ 6 & 3 & 7 \end{vmatrix}$$

• •7(• 14• 24) • •(• 1)(70• 48) • •0(30 • •12)

• •7(• 38) • •1(22) • •0

• •• 266 • •22 • •• 244

$$x_1 = \frac{\Delta x_1}{\Delta} = \frac{-61}{-61} = 1$$